Cross-Cohort Validation of Clinical Prediction Models in First-Episode Psychosis

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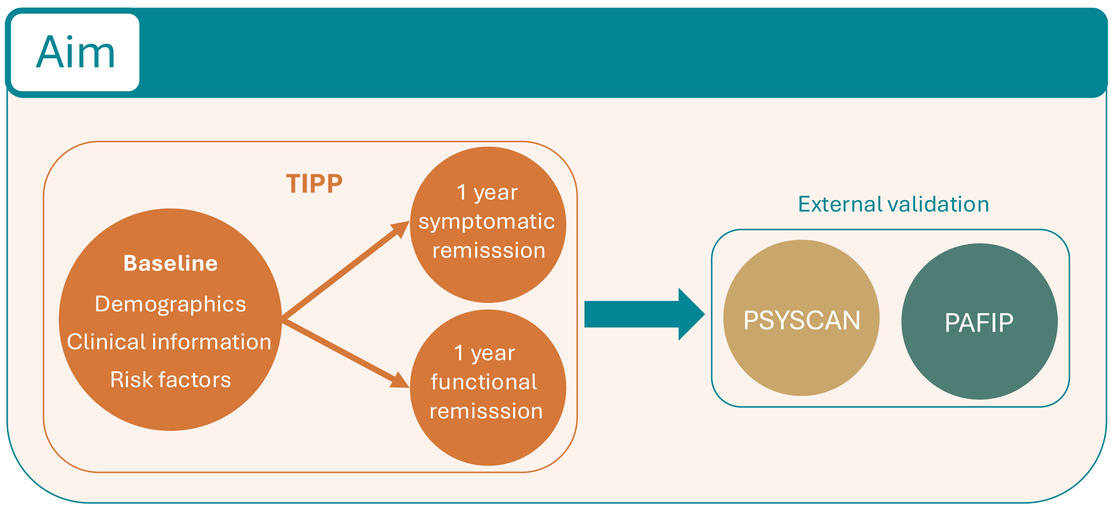
Introduction

People presenting with first-episode psychosis (FEP) have heterogenous outcomes[1]. Accurate prediction of individual outcome in FEP could facilitate early intervention to change the clinical trajectory and improve prognosis.

Several prediction models have been developed to estimate likely treatment outcomes, but very few have been externally validated [2]. External validation is essential because it provides a rigorous test of whether the information learned by a model truly distinguishes between patients who later improve and those who do not [3]. This remains a major challenge, as clinical services often differ in procedures, assessments, and patient characteristics, creating inherent differences across samples.

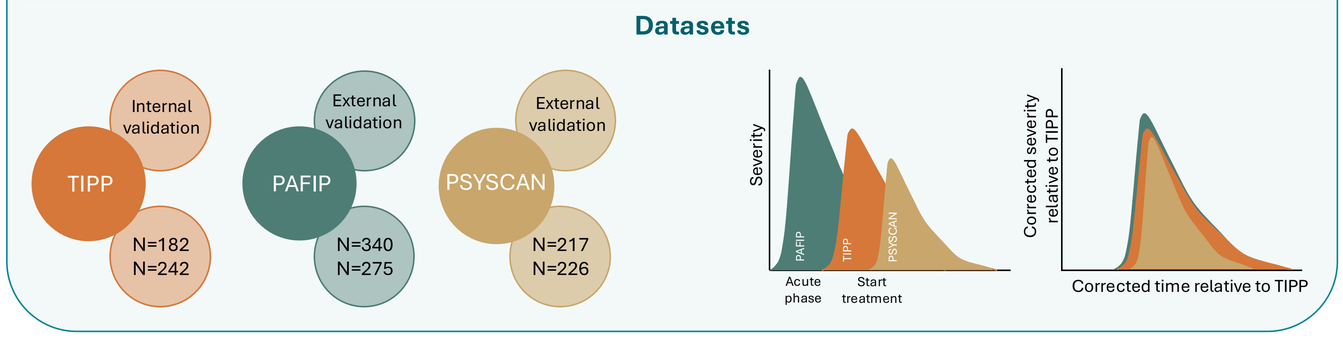
Previous studies have used a range of predictors[2]. Routinely acquired clinical data may be the most promising for implementation, as it is already available in practice and does not require additional costly assessments.

Aim

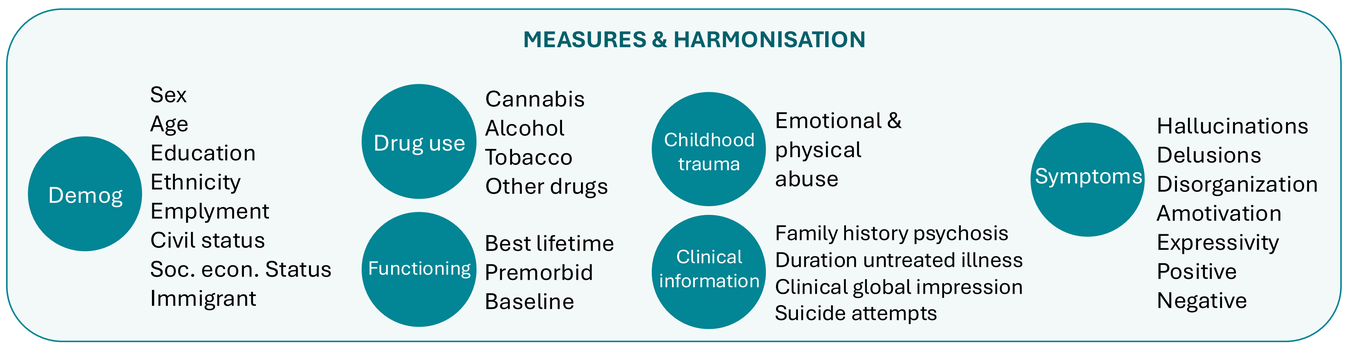


Methods

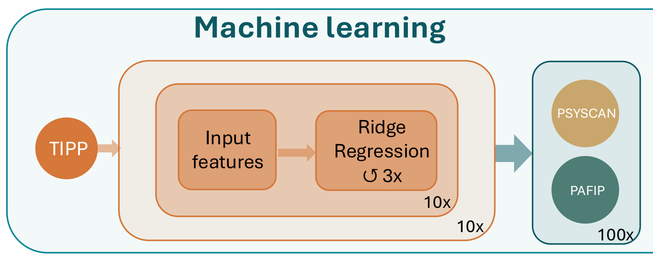
Datasets



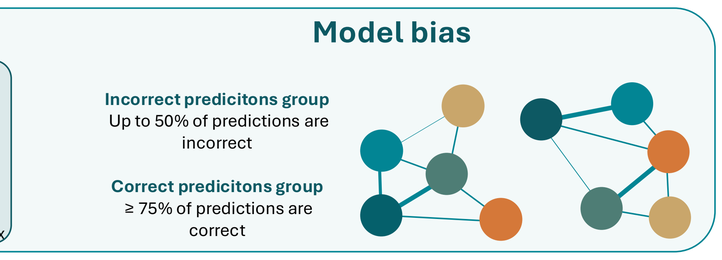
MEASURES & HARMONISATION



Machine learning

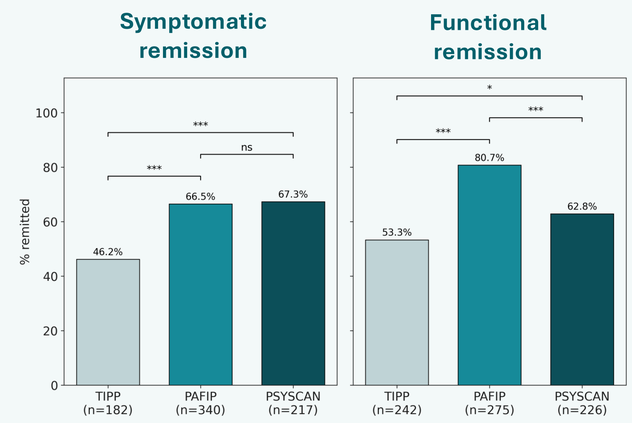


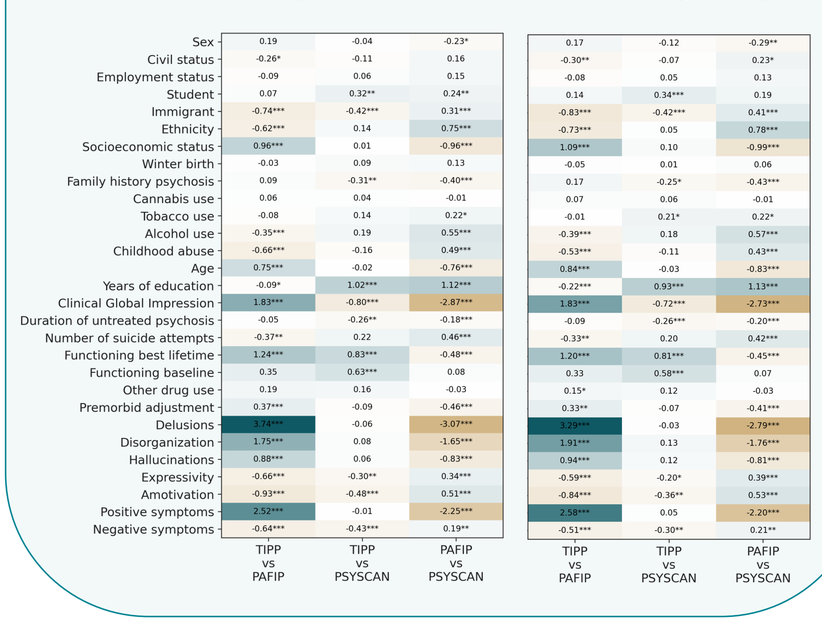
Model bias



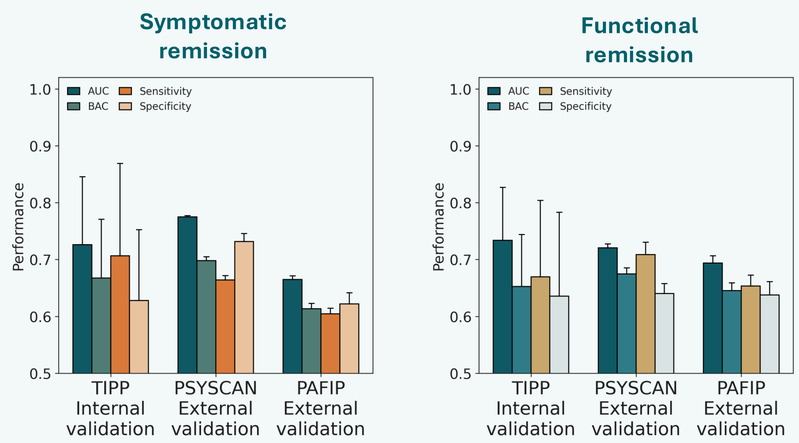
Results

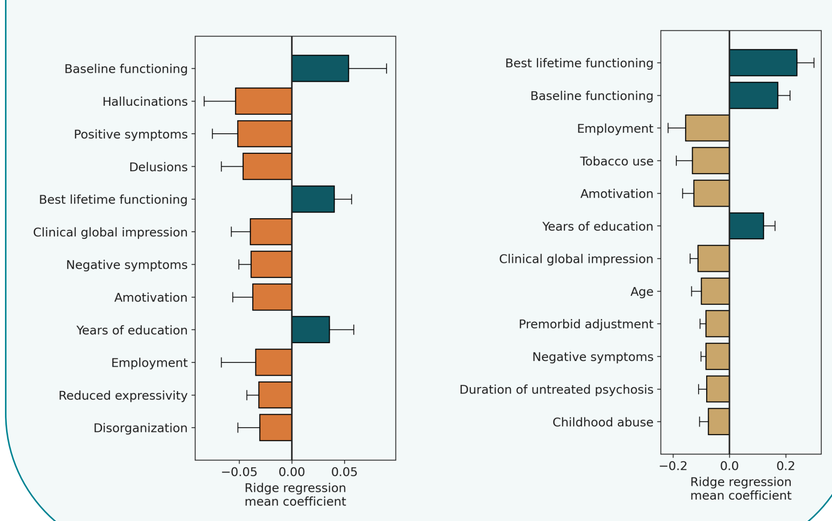
1. Sample charateristics





2. Model performance and most informative features

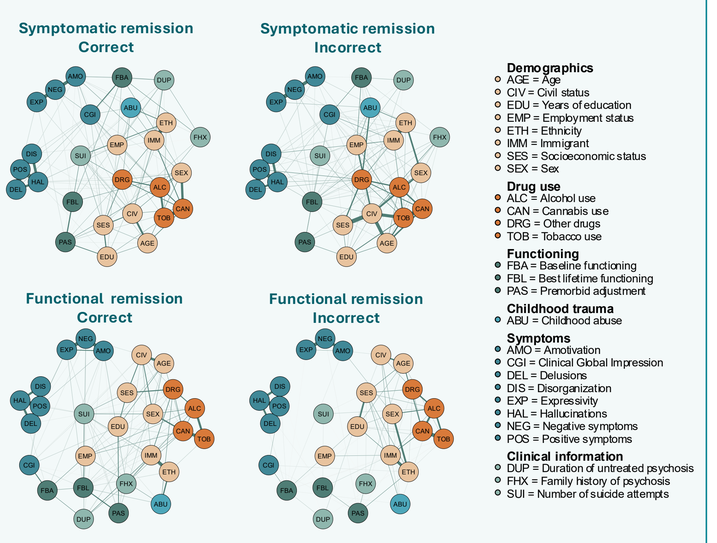




3. Model bias

	Group (n)	Total n (%)	TIPP n (%)	PSYSCAN n (%)	PAFIP n (%)	% of features sig ≠
Symptomatic remission	Correct	459 (63.7)	116 (65.2)	145 (68.3)	198 (59.8)	22.2% inc > corr symptoms
	Incorrect	262 (36.3)	62 (34.8)	67 (32.6)	133 (40.2)	
Functional remission	Correct	455 (63.1)	150 (62.8)	144 (66.1)	161 (62.4)	0%
	Incorrect	255 (35.9)	89 (37.2)	74 (33.9)	97 (37.6)	

Sample size and univariate comparison between the Correct and Incorrect prediction groups on models' input features adjuted for cohort and multiple comparisons.



Outcome	Network structure invariance	Global strength invariance	Strength correlation stability
Symptomatic remission	M = 0.6 p=0.038	S = 3.8 p = 0.221	CS_{cor} = 0.52 CS_{inc} = 0.60
Functional remission	M = 0.5 p=0.175	S = 3.1 p = 0.299	CS_{cor} = 0.67 CS_{inc} = 0.60

Correct vs Incorrect networks.

Conclusion

- External validation remains rare in FEP prediction models → this study provides validation of a 1-year outcome prediction model across two independent cohorts.
- Models trained in TIPP showed good transportability to both external datasets. For symptomatic remission, the model achieved an AUC of 0.72 in internal and 0.78 and 0.66 in external validations. For functional remission, the model achieved an AUC of 0.73 in internal validation and 0.72 and 0.69 in external validation.
- Symptomatic remission was driven more by symptom severity, whereas functional remission was driven by psychosocial and functioning-related variables.
- There were no differences in clinical profiles between patients with correct and incorrect predictions for functional remission → no apparent bias.
- For symptomatic remission, patients with more severe symptoms had lower rate of correct predictions → bias towards less severe patients.

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